Analysis of Risk Factors for Delayed Graft Function After Kidney Transplantation

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ABSTRACT

Background. Delayed graft function (DGF) is a serious complication associated with worsening outcomes in kidney transplantation. To facilitate DGF risk reduction, this study aimed to identify the incidence and modifiable risk factors of this condition in kidney transplant patients.

Methods. This retrospective chart review included 220 patients who underwent kidney transplants between 2012 and 2021 at our kidney transplant center. Delayed graft function was defined as the requirement of hemodialysis within a week of transplantation. Clinical data from patients with DGF and those without this condition were compared to identify risk factors of DGF.

Results. Of 205 eligible patients, 20 (9.76%) developed DGF. In the univariate analysis, high hemoglobin level, deceased-donor type, and longer warm and cold ischemic times were significantly associated with DGF ($P < .05$). In the variable selection in logistic regression analysis, high hemoglobin level, with a cutoff value of 11.35 g/dL, and deceased-donor transplants were associated with higher DGF incidence ($P < .05$ for both factors).

Conclusions. Our findings newly demonstrated that DGF occurred more frequently in patients with hemoglobin level >11.35 g/dL. As such, improvement in kidney transplantation outcomes could be achieved by reducing this modifiable risk factor.

DELAYED graft function (DGF) is defined as the requirement for hemodialysis at least once within the first postoperative week because of the occurrence of symptoms such as oliguria and uremia, despite of kidney transplantation [1]. Delayed graft function, a common post-transplant complication, occurs in at least 5% to 25% of all kidney transplant recipients [2]. However, the presence of DGF lengthens the hospital stay and increases medical costs. Ultimately, this complication affects graft outcomes and increases morbidity and mortality [3]. Efforts to improve donor and recipient management and treatment modalities have not resulted in a reduction in the overall rate of this disorder [4]. In this study, DGF-related factors analysis was performed to identify DGF risk factors.

MATERIALS AND METHODS

Study Design

This retrospective chart review analyzed 220 patients who underwent kidney transplants at our kidney transplant center between March 2012 and November 2021. This study was approved by the Institutional Review Board of the Pusan National University Yangsan Hospital. The study was conducted in accordance with the guidelines from the Declaration of Helsinki. The exclusion criteria were as follows: graft loss due to technical failure, multiple organ transplants, hyperacute rejection, patients aged <18 years and those with incomplete or unavailable charts. On the day of transplantation surgery, blood samples were collected from each patient in the early morning after fasting for ≥6 hours. Among the patients who underwent kidney transplantation, the clinical data from those with and without DGF were collected. Parameters with possible influence on DGF were compared between the DGF and non-DGF groups. The following information was analyzed: recipient sex, age, body weight, height, body mass index, ABO blood type, hemoglobin level at kidney transplantation, warm and cold ischemia times, donor type, mismatch of human leukocyte antigen, graft kidney

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weights, panel reactive antibody, renal replacement therapy modality before transplantation, intact parathyroid hormone, brain natriuretic peptide, erythropoietin, neutrophil gelatinase-associated lipocalin, and immunosuppression protocol.

Statistical Analysis

Depending on the data type, demographic information was summarized using frequency (%) or mean (SD). The Fisher's exact test and χ^2 test were used to compare the categorical variables, and an independent t test was used to compare the continuous variables. The risk factors for graft loss and DGF were assessed using a logistic regression model. To identify the appropriate hemoglobin cutoff value with optimal specificity and sensitivity for the prediction of DGF, receiver operating characteristic (ROC) curves were used. We also calculated the positive and negative predictive values for the cutoff values of hemoglobin for DGF prediction. Statistical significance was defined as $\leq P \leq .05$. The IBM SPSS Statistics 26.0 software (Armonk, NY) was used to conduct statistical analyses.

RESULTS

Among 220 patients, 205 (121 men and 84 women) met the eligibility criteria of the study (Table 1). Delayed graft function was present in 20 of 205 (9.76%) patients. The average recipient age was 53.0 years, and 55% of the patients were male (n = 11). The risk factors for DGF development were assessed using a logistic regression model (Table 2). There were no significant differences in age ($P = .071$), sex ($P = .700$), and weight ($P = .563$) between DGF and non-DGF patients. Patients with DGF experienced average warm and cold ischemia times of

Table 1. Patient Characteristics Based on the Presence of Delayed Graft Function

			DGF	
Variables	Characteristics	N (%) / M ± SD	No N (%) / M ± SD	Yes N (%) / M ± SD
Sex of recipient	Male	121 (59)	110 (59.5)	11 (55)
	Female	84 (41)	75 (40.5)	9 (45)
Age (y)		48.43 ± 11.75	47.94 ± 11.95	53.00 ± 8.68
Body weight (kg)		64.92 ± 14.18	64.73 ± 14.49	66.66 ± 11.10
Height (m)		1.66 ± 0.09	1.66 ± 0.09	1.67 ± 0.08
BMI (kg/m^2)		23.40 ± 4.07	23.34 ± 4.15	23.95 ± 3.27
CKD	Known	141 (68.8)	124 (67)	17 (85)
	Unknown	64 (31.2)	61 (33)	3 (15)
Hemoglobin		10.61 ± 1.67	10.49 ± 1.62	11.70 ± 1.71
Intact PTH		233.34 ± 190.07	233.66 ± 192.24	230.43 ± 173.20
Type of RRT	Hemodialysis	136 (66.3)	117 (63.2)	19 (95)
	Peritoneal dialysis	27 (13.2)	26 (14.1)	1 (5)
	Pre-emptive	42 (20.5)	42 (22.7)	0 (0)
Type of ABO	Match	181 (88.3)	162 (87.6)	19 (95)
	Mismatch	24 (11.7)	23 (12.4)	1 (5)
Mismatch of HLA-A (0.1,2)		1.10 ± 0.67	1.09 ± 0.68	1.20 ± 0.62
Mismatch of HLA-B (0.1,2)		1.34 ± 0.65	1.32 ± 0.66	1.55 ± 0.51
Mismatch of HLA-DR (0.1,2)		1.19 ± 0.68	1.16 ± 0.68	1.40 ± 0.60
Preoperative PRA	Negative	102 (49.8)	94 (50.8)	8 (40)
	Positive	103 (50.2)	91 (49.2)	12 (60)
Preoperative PRA class I %		19.18 ± 27.32	18.27 ± 27.33	26.08 ± 27.42
Preoperative PRA class II %		30.04 ± 29.89	28.94 ± 30.01	38.25 ± 28.86
Preoperative DSA	No	57 (55.3)	50 (54.9)	7 (58.3)
	Yes	46 (44.7)	41 (45.1)	5 (41.7)
Type of donor	Living	98 (47.8)	95 (51.4)	3 (15)
	Deceased	107 (52.2)	90 (48.6)	17 (85)
Sex of donor	Male	113 (55.1)	99 (53.5)	14 (70)
	Female	92 (44.9)	86 (46.5)	6 (30)
Warm ischemic time (min)		36.41 ± 19.46	35.20 ± 17.73	47.60 ± 29.65
Cold ischemic time (min)		111.04 ± 107.16	104.76 ± 106.65	169.20 ± 95.94
Graft kidney weight (g)		188.83 ± 40.27	187.46 ± 40.15	201.55 ± 40.14
Induction therapy	ATG	41 (20)	36 (19.5)	5 (25)
	Basiliximab	164 (80)	149 (80.5)	15 (75)
EPO		92.67 ± 137.17	89.84 ± 133.53	111.83 ± 168.79
NGAL		629.32 ± 382.53	634.02 ± 389.78	594.09 ± 344.08
BNP		253.13 ± 572.38	246.35 ± 588.79	315.88 ± 413.19

Data were presented as absolute number and percentage, n (%), or mean ± SD, (M ± SD).

ATG, antithymocyte globulin; BMI, body mass index; BNP, brain natriuretic peptide; CKD, chronic kidney disease; DGF, delayed graft function; DSA, donor-specific antibodies; EPO, erythropoietin; HLA, human leukocyte antigen; M, mean; NGAL, neutrophil gelatinase-associated lipocalin; PRA, panel reactive antibody; PTH, parathyroid hormone; RRT, renal replacement therapy.

Table 2. Multivariate Logistic Regression Analysis for Predicting Delayed Graft Function

		Univariate model		Multivariate model	
Variables	Characteristics	Odds Ratio (95% CI)	*P* value	Odds Ratio (95% CI)	*P* value
Sex of recipient	Male				
	Female	1.200 (0.463-3.039)	.700	1.557 (0.301-8.063)	.598
Age (y)		1.043 (0.999-1.095)	.071	1.046 (0.982-1.113)	.160
Body weight (kg)		1.009 (0.977-1.040)	.563	0.832 (0.525-1.320)	.435
Height (m)		2.205 (0.011-45.986)	.769	1.218 (0.823-1.803)	.324
BMI (kg/m^2)		1.036 (0.925-1.155)	.525	1.822 (0.483-6.877)	.376
Cause of ESRD	Known				
	Unknown	0.359 (0.101-1.271)	.112	0.238 (0.055-1.018)	.053
Hemoglobin		1.607 (1.190-2.241)	.003	1.504 (1.061-2.132)	.022
Intact PTH		1.000 (0.997-1.002)	.942	1.000 (0.047-5.193)	.558
Type of RRT	Hemodialysis				
	Peritoneal dialysis	0.237 (0.013-1.224)	.170	0.495 (0.047-5.193)	.558
	Pre-emptive	0.311 (0.255-1.595)	.992		
Type of ABO	Match				
	Mismatch	0.371 (0.020-1.924)	.344		
Mismatch of HLA-A (0.1,2)		1.292 (0.647-2.646)	.473		
Mismatch of HLA-B (0.1,2)		1.833 (0.862-4.290)	.135		
Mismatch of HLA-DR (0.1,2)		1.742 (0.858-3.758)	.138		
Preoperative PRA	Negative				
	Positive	1.549 (0.612-4.120)	.361		
Preoperative PRA class I %		1.009 (0.988-1.029)	.355		
Preoperative PRA class II %		1.010 (0.990-1.029)	.313		
Preoperative DSA	No				
	Yes	0.871 (0.242-2.931)	.824		
Type of donor	Living				
	Deceased	5.981 (1.929-26.239)	.005	2.649 (0.557-12.604)	.221
Sex of donor	Male				
	Female	0.493 (0.169-1.288)	.166	0.989 (0.269-3.643)	.987
Warm ischemic time (min)		1.024 (1.005-1.043)	.012	1.016 (0.995-1.037)	.129
Cold ischemic time (min)		1.005 (1.001-1.008)	.015	1.001 (0.996-1.006)	.696
Graft kidney weight (g)		1.008 (0.997-1.020)	.140	1.003 (0.989-1.018)	.637
Induction therapy	ATG				
	Basiliximab	0.725 (0.261-2.345)	.558	1.678 (0.397-7.101)	.482
EPO		1.001 (0.995-1.006)	.671		
NGAL		1.000 (0.997-1.002)	.780		
BNP		1.000 (0.998-1.001)	.747		

ATG, antithymocyte globulin; BMI, body mass index; BNP, brain natriuretic peptide; DSA, donor-specific antibodies; EPO, erythropoietin; ESRD, end stage renal disease; HLA, human leukocyte antigen; NGAL, neutrophil gelatinase-associated lipocalin; PRA, panel reactive antibody; PTH, parathyroid hormone; RRT, renal replacement therapy.

36.41 ± 19.46 (mean ± SD) minutes and 111.04 ± 107.16 minutes, respectively, both of that were significantly higher than values of the non-DGF group. A total of 98 (47.8%) patients underwent living-donor renal transplantation, whereas 107 (52.2%) patients underwent deceased-donor kidney transplantation. Delayed graft function occurred more frequently in patients who received deceased-donor kidneys (n = 17) than in those transplanted with living-donor donor kidneys (n = 3) ($P <$ .05). Patients with DGF had higher hemoglobin levels than those without DGF ($P <$ .05). Among the 20 patients who developed DGF, 19 underwent ABO-type matched transplantation and 1 underwent ABO-type mismatched transplantation. However, there were no statistically significant association between the ABO blood type and DGF. There was also no significant difference between the degree of human leukocyte antigen mismatch and DGF (P = 0.081). All patients received basiliximab or antithymocyte globulin as the induction therapy. After the induction therapy, triple immunosuppressive drugs (prednisolone, tacrolimus, and mycophenolic acid) were administered as the maintenance therapy. There were no statistically significant difference in DGF occurrence between patients receiving the induction therapy and those with the maintenance therapy. There were also no significant associations between DGF with the pretransplant dialysis method or transplant kidney size. In variable selection in logistic regression models, significant factors for DGF were hemoglobin level (odds ratio: 1.47, 95% CI: 1.04-2.08, P = .029) and deceased-donor type (odds ratio: 4.10, 95% CI: 1.128-14.958, P = .032) (Table 3). The ROC curve was used to assess the prognostic value of hemoglobin level for DGF and the result is shown in Fig 1 (area under the curve = 0.69; 95% CI: 0.616-0.736; P = .003). In addition, the optimal cutoff value of hemoglobin for DGF was

Table 3. Variable Selection in Logistic Regression Model

Variables	Characteristics	Odds Ratio (95% CI)	*P* value
Hemoglobin		1.474 (1.042-2.086)	.029
Type of donor	Living		
	Deceased	4.107 (1.128-14.958)	.032

11.35 g/dL, with a specificity of 70.3% and sensitivity of 65%. The positive and negative predictive values were 19% and 94%, respectively.

DISCUSSION

In transplanted kidneys, the DGF results from by hypoperfusion-induced cellular damage, which is worsened by ischemia-reperfusion injury [5]. Ischemic harm to a kidney graft is the sum of all likely temporary warm ischemic time before or during the removal process from the donor and the cold ischemia associated with preservation and storage [6]. Long periods of warm or cold ischemia may enhance DGF occurrence and trigger some preprogrammed processes in the kidneys, resulting in chronic and progressive lesions [7,8]. Although increased warm and cold ischemic times were observed in DGF patients in our center study, these parameters did not show significant associations with DGF in the multivariate analysis. According to a recent study of 846 patients who underwent kidney transplantation, the average time required for ischemia was 1080 minutes, which was markedly longer than that of our center [9]. The risk of DGF increases by 23% with every 6 hours of ischemia time, and the incidence of DGF increases substantially with increasing ischemic time [10]. Despite the importance of ischemic time as a risk factor for DGF, the absence of its association with DGF in our multivariate analysis might be related to the shorter ischemic time in our study compared to those from other centers. The shorter ischemic period might result from a more abbreviated transport distance of organs due to the geographic characteristics of Korea. In this regard, deceased-donor kidney transplantation recipients often receive organs from deceased donors managed by our hospital. As the kidney transplantation ratio of living donors in our study was approximately half, the cold ischemia time can be shortened, which did not offset other DGF risks or cause any significant harms. Donor type and high hemoglobin concentration before kidney transplantation were significant risk factors for DGF in variable selection logistic regression analysis. This higher incidence of DGF in deceased-donor kidney transplantation in comparison to that of living-donor kidney transplantation is consistent with findings from several previous studies [11−14]. The goal of this study was to examine novel risk factors that can cause DGF. To the best of our knowledge, low hemoglobin level is an important factor in DGF development. The association between low hemoglobin

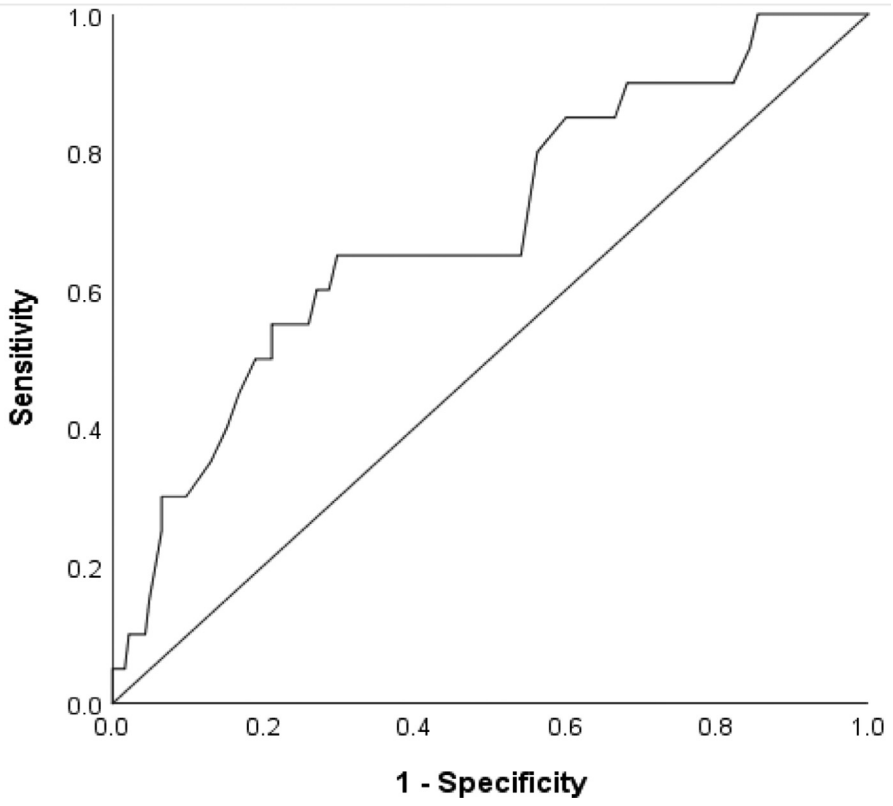


Fig. 1. Receiver operating characteristic (ROC) curve for hemoglobin level to predict the DGF development. ROC curve of hemoglobin with 95% confidence intervals. The area under curve (AUC) is 0.690 (from 0.616 to 0.736), and the cut-off value is 11.35 g/dL with the sensitivity of 65.0% and specificity of 70.3%.

concentration and DGF is consistent with findings from other studies [15]. Anemia is a common symptom of erythropoietin deficiency, malnutrition, and blood loss due to dialysis in patients with chronic renal failure who usually require transplantation. Hypoperfusion occurs when oxygen delivery is insufficient, and management of anemia is essential because hemoglobin is an important component of oxygen delivery. A recent prospective cohort study showed that anemia at baseline was associated with adverse outcomes, including mortality and graft loss [15]. In theory, high blood hemoglobin level can prevent ischemia-reperfusion injury after transplantation. Among the factors contributing to the induction of DGF, the maintenance of renal blood flow and oxygen supply by proper control of hemoglobin level could help prevent DGF. However, no studies have defined an appropriate range of hemoglobin concentration. Our evaluation of the association between DGF and hemoglobin level in kidney transplant patients show that, with the ROC curve, the appropriate cutoff level for hemoglobin was 11.35 g/dL. Because adequate hemoglobin level can reduce post-transplant ischemia and prevent ischemia-reperfusion injury, the influence of blood hemoglobin level on the incidence of DGF has a U-shaped effect. However, complications such as graft thrombosis may arise if hemoglobin level increases above a specific threshold. Doppler ultrasonography in patients with DGF often results in an elevated renal resistive index, indicative of impaired blood flow to the kidney. Although this observation may support the presence of thrombosis [16], more investigation is required to elucidate the detailed underlying mechanism. Our study had the limitation of being a retrospective study. In this regard, it cannot account for all possible relevant factors and may have some bias. However, we evaluated kidney donor types and ischemia times with particular reference to the hemoglobin level. Currently, there is no effective prevention strategy and treatment for DGF. Because the donor population is immutable, keeping the ischemia time as short as possible and maintaining hemoglobin level within the appropriate range may help prevent DGF.

In conclusion, identifying modifiable predictors of DGF is important to prevent DGF by facilitating early intervention after kidney transplantation and ultimately minimizing such potential risk factors.

DATA AVAILABILITY

Data will be made available on request.

DISCLOSURE

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